

Question:

Patient (PatientID, Name, Gender, DOB, Phone, Address)

Doctor (DoctorID, Name, Specialty, Phone)

Appointment (AppointmentID, PatientID, DoctorID, AppointmentDate, Status)

Prescription (PrescriptionID, AppointmentID, Diagnosis, PrescribedMedicines)

Medicine (MedicineID, Name, Stock, Price)

Bill (BillID, AppointmentID, Amount, BillingDate, PaymentStatus)

Insert these in the database

Patient:

PatientID	Name	Gender	DOB	Phone	Address
1	Alice Rahman	Female	1995-06-12	01711112222	Dhaka, Bangladesh
2	Imran Hossain	Male	1988-03-25	01822223333	Chittagong, Bangladesh
3	Rafiq Ahmed	Male	1972-11-10	01933334444	Sylhet, Bangladesh

Doctor:

DoctorID	Name	Speciality	Phone
101	Dr. Sara Islam	Cardiology	01755556666
102	Dr. Tanvir Hasan	Neurology	01866667777
103	Dr. Nusrat Jahan	General Medicine	01977778888

Appointment:

AppointmentID	PatientID	DoctorID	AppointmentDate	Status
201	1	101	2025-06-10 10:00:00	Completed
202	2	102	2025-06-12 11:30:00	Scheduled
203	3	103	2025-06-15 09:00:00	Cancelled

Prescription:

PrescriptionID	AppointmentID	Diagnosis	PrescribedMedicines
301	201	High blood pressure	Sumatriptan 50mg, when needed
302	202	Migraine	Sumatriptan 50mg, when needed

Medicine:

MedicineID	Name	Stock	Price
401	Amlodipine 5mg	150	10.00
402	Sumatriptan 50mg	80	35.00
403	Paracetamol 500mg	500	2.50

Bill:

BillID	AppointmentID	Amount	BillingDate	PaymentStatus
501	201	500.00	2025-06-10	Paid
502	202	750.00	2025-06-12	Unpaid

Queries:

1. List the names and phone numbers of all doctors who specialize in "Cardiology".
2. Show the names and appointment dates of all patients whose appointment status is "Scheduled".
3. List the names of patients who have received a prescription containing the medicine "Amlodipine 5mg".
4. Find the names of doctors who have at least one appointment marked as "Completed".
5. Display each patient's name along with the amount and payment status of their bill.
6. Retrieve the names of patients who live in "Sylhet" and had their appointment cancelled.
7. Count how many appointments each doctor has, and show the doctor's name with the count.
8. List the names of all patients who do not have any unpaid bills.
9. Show the total bill amount generated for all appointments before '2025-06-12'.

Answer:

Creating Database:

```
CREATE TABLE Patient (  
    PatientID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    Gender VARCHAR(10),  
    DOB DATE,  
    Phone VARCHAR(15),  
    Address TEXT  
);
```

```
CREATE TABLE Patient ( PatientID INT PRIMARY KEY, Name VARCHAR(100), Gender VARCHAR(10), DOB DATE, Phone VARCHAR(15), Address TEXT );
```

```
CREATE TABLE Doctor (  
    DoctorID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    Specialty VARCHAR(100),  
    Phone VARCHAR(15)  
);
```

```
CREATE TABLE Doctor ( DoctorID INT PRIMARY KEY, Name VARCHAR(100), Specialty VARCHAR(100), Phone VARCHAR(15) );
```

```
CREATE TABLE Appointment (  
    AppointmentID INT PRIMARY KEY,  
    PatientID INT,  
    DoctorID INT,  
    AppointmentDate DATETIME,  
    Status VARCHAR(20),  
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),  
    FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID)  
);
```

```
CREATE TABLE Appointment ( AppointmentID INT PRIMARY KEY, PatientID INT, DoctorID INT, AppointmentDate DATETIME, Status VARCHAR(20), FOREIGN KEY (PatientID) REFERENCES Patient(PatientID), FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID) );
```

```
CREATE TABLE Prescription (  
    PrescriptionID INT PRIMARY KEY,  
    AppointmentID INT,  
    Diagnosis TEXT,  
    PrescribedMedicines TEXT,  
    FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID)  
);
```

```
CREATE TABLE Prescription ( PrescriptionID INT PRIMARY KEY, AppointmentID INT, Diagnosis TEXT, PrescribedMedicines TEXT, FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID) );
```

```
CREATE TABLE Medicine (  
    MedicineID INT PRIMARY KEY,  
    Name VARCHAR(100),  
    Stock INT,  
    Price DECIMAL(10,2)  
);
```

```
CREATE TABLE Medicine ( MedicineID INT PRIMARY KEY, Name VARCHAR(100), Stock INT, Price DECIMAL(10,2) );
```

```
CREATE TABLE Bill (  
    BillID INT PRIMARY KEY,  
    AppointmentID INT,  
    Amount DECIMAL(10,2),  
    BillingDate DATE,  
    PaymentStatus VARCHAR(20),  
    FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID)  
);
```

```
CREATE TABLE Bill ( BillID INT PRIMARY KEY, AppointmentID INT, Amount DECIMAL(10,2), BillingDate DATE, PaymentStatus VARCHAR(20), FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID) );
```

Insertions:

INSERT INTO Patient VALUES

(1, 'Alice Rahman', 'Female', '1995-06-12', '01711112222', 'Dhaka, Bangladesh'),
(2, 'Imran Hossain', 'Male', '1988-03-25', '01822223333', 'Chittagong, Bangladesh'),
(3, 'Rafiq Ahmed', 'Male', '1972-11-10', '01933334444', 'Sylhet, Bangladesh');

✓ 3 rows inserted. (Query took 0.0008 seconds.)

```
INSERT INTO Patient VALUES (1, 'Alice Rahman', 'Female', '1995-06-12', '01711112222', 'Dhaka, Bangladesh'), (2, 'Imran Hossain', 'Male', '1988-03-25', '01822223333', 'Chittagong, Bangladesh'), (3, 'Rafiq Ahmed', 'Male', '1972-11-10', '01933334444', 'Sylhet, Bangladesh');
```

INSERT INTO Doctor VALUES

(101, 'Dr. Sara Islam', 'Cardiology', '01755556666'),
(102, 'Dr. Tanvir Hasan', 'Neurology', '01866667777'),
(103, 'Dr. Nusrat Jahan', 'General Medicine', '01977778888');

✓ 3 rows inserted. (Query took 0.0010 seconds.)

```
INSERT INTO Doctor VALUES (101, 'Dr. Sara Islam', 'Cardiology', '01755556666'), (102, 'Dr. Tanvir Hasan', 'Neurology', '01866667777'), (103, 'Dr. Nusrat Jahan', 'General Medicine', '01977778888');
```

INSERT INTO Appointment VALUES

(201, 1, 101, '2025-06-10 10:00:00', 'Completed'),
(202, 2, 102, '2025-06-12 11:30:00', 'Scheduled'),
(203, 3, 103, '2025-06-15 09:00:00', 'Cancelled');

✓ 3 rows inserted. (Query took 0.0011 seconds.)

```
INSERT INTO Appointment VALUES (201, 1, 101, '2025-06-10 10:00:00', 'Completed'), (202, 2, 102, '2025-06-12 11:30:00', 'Scheduled'), (203, 3, 103, '2025-06-15 09:00:00', 'Cancelled');
```

INSERT INTO Prescription VALUES

(301, 201, 'High blood pressure', 'Amlodipine 5mg, once daily'),
(302, 202, 'Migraine', 'Sumatriptan 50mg, when needed');

✓ 2 rows inserted. (Query took 0.0010 seconds.)

```
INSERT INTO Prescription VALUES (301, 201, 'High blood pressure', 'Amlodipine 5mg, once daily'), (302, 202, 'Migraine', 'Sumatriptan 50mg, when needed');
```

INSERT INTO Medicine VALUES

(401, 'Amlodipine 5mg', 150, 10.00),
(402, 'Sumatriptan 50mg', 80, 35.00),
(403, 'Paracetamol 500mg', 500, 2.50);

✓ 3 rows inserted. (Query took 0.0009 seconds.)

```
INSERT INTO Medicine VALUES (401, 'Amlodipine 5mg', 150, 10.00), (402, 'Sumatriptan 50mg', 80, 35.00), (403, 'Paracetamol 500mg', 500, 2.50);
```

INSERT INTO Bill VALUES

(501, 201, 500.00, '2025-06-10', 'Paid'),

(502, 202, 750.00, '2025-06-12', 'Unpaid');

✓ 2 rows inserted. (Query took 0.0009 seconds.)

```
INSERT INTO Bill VALUES (501, 201, 500.00, '2025-06-10', 'Paid'), (502, 202, 750.00, '2025-06-12', 'Unpaid');
```

Queries:

1. List the names and phone numbers of all doctors who specialize in "Cardiology".

SELECT Name, Phone

FROM Doctor

WHERE Specialty = 'Cardiology';

Name	Phone
Dr. Sara Islam	01755556666

2. Show the names and appointment dates of all patients whose appointment status is "Scheduled".

SELECT Patient.Name, Appointment.AppointmentDate

FROM Patient

JOIN Appointment ON Patient.PatientID = Appointment.PatientID

WHERE Appointment.Status = 'Scheduled';

Name	AppointmentDate
Imran Hossain	2025-06-12 11:30:00

3. List the names of patients who have received a prescription containing the medicine "Amlodipine 5mg"

SELECT DISTINCT Patient.Name

FROM Patient

JOIN Appointment ON Patient.PatientID = Appointment.PatientID

JOIN Prescription ON Appointment.AppointmentID = Prescription.AppointmentID
WHERE PrescribedMedicines LIKE '%Amlodipine 5mg%';

Name

Alice Rahman

4. Find the names of doctors who have at least one appointment marked as "Completed".

SELECT DISTINCT Doctor.Name

FROM Doctor

JOIN Appointment ON Doctor.DoctorID = Appointment.DoctorID

WHERE Appointment.Status = 'Completed';

Name

Dr. Sara Islam

5. Display each patient's name along with the amount and payment status of their bill.

SELECT Patient.Name, Bill.Amount, Bill.PaymentStatus

FROM Patient

JOIN Appointment ON Patient.PatientID = Appointment.PatientID

JOIN Bill ON Appointment.AppointmentID = Bill.AppointmentID;

Name	Amount	PaymentStatus
Alice Rahman	500.00	Paid
Imran Hossain	750.00	Unpaid

6. Retrieve the names of patients who live in "Sylhet" and had their appointment cancelled.

SELECT DISTINCT Patient.Name

FROM Patient

JOIN Appointment ON Patient.PatientID = Appointment.PatientID

WHERE Patient.Address LIKE '%Sylhet%' AND Appointment.Status = 'Cancelled';

Name

Rafiq Ahmed

7. Count how many appointments each doctor has, and show the doctor's name with the count.

```
SELECT Doctor.Name, COUNT(Appointment.AppointmentID) AS TotalAppointments
FROM Doctor
LEFT JOIN Appointment ON Doctor.DoctorID = Appointment.DoctorID
GROUP BY Doctor.DoctorID, Doctor.Name;
```

Name	TotalAppointments
Dr. Sara Islam	1
Dr. Tanvir Hasan	1
Dr. Nusrat Jahan	1

8. List the names of all patients who do not have any unpaid bills.

```
SELECT DISTINCT Patient.Name
FROM Patient
JOIN Appointment ON Patient.PatientID = Appointment.PatientID
JOIN Bill ON Appointment.AppointmentID = Bill.AppointmentID
WHERE Bill.PaymentStatus != 'Unpaid';
```

Name
Alice Rahman

9. Show the total bill amount generated for all appointments before '2025-06-12'.

```
SELECT SUM(Amount) AS TotalBill
FROM Bill
JOIN Appointment ON Bill.AppointmentID = Appointment.AppointmentID
WHERE Appointment.AppointmentDate < '2025-06-12';
```

TotalBill
500.00